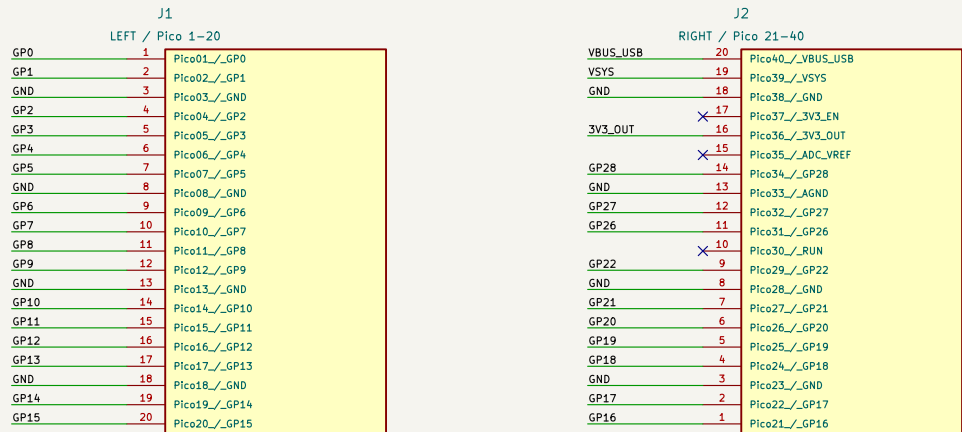


MAKE MUSIC – PROVISIONAL PICO CARRIER

Editable KiCad 10 schematic | Module order, parts, power and mechanical fit require primary integration

REMOVABLE PICO: TOP VIEW, USB AT TOP

Selected Pico H SC0917: original RP2040, fitted male headers. Socket pitch2.54mm / rows17.78mm; fit trial pending.



J1 pin 1 = Pico 1 (USB end), J2 pin 1 = Pico 21 (opposite USB).

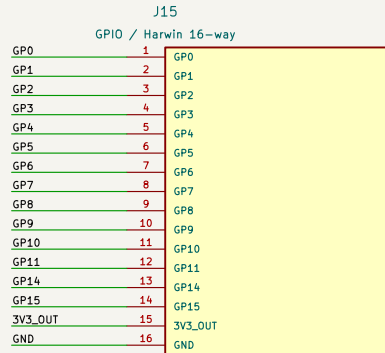
J2 pad 20 = Pico 40 at USB end: rotate J2 footprint 180 degrees relative to J1 in placement.

NC: Pico 30 RUN, 35 ADC_VREF, 37 3V3_EN, On-module defaults retained.

Pico 33 AGND tied to GND for digital touch use. GP23–25/29 are internal, not expansion.

SSQ candidate: 1.02 mm drills / 1.52 mm pads. Verify finished holes and male–header fit.

FREE GPIO EXPANSION – CARRIER ORDER DEFINED



GP0/GP1: optional I2C SDA/SCL.

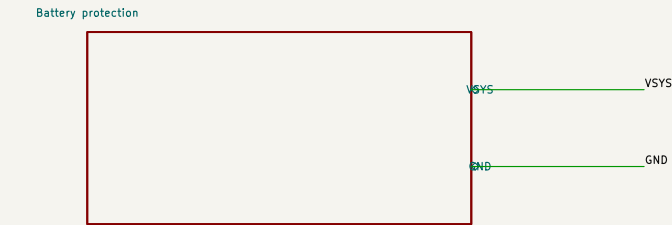
No I2C pull-ups fitted until bus/device choice.

REVIEW ONLY – NOT FOR FABRICATION

ERC checks carrier wiring only. Cell PCM, module electronics and assembled fault/load/fit behavior remain unverified. See power sheet.

Sources: Raspberry Pi original Pico datasheet, Figure 2 pin allocation; see ../reviews/schematic.md.

FUSED BATTERY POWER – SHEET 2



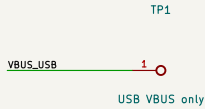
File: battery_power.kicad_sch

H1 JST harness: circuit1 protected positive, circuit2 GND.

F1 + U2 replace the old carrier series D1.

Switch OFF isolates battery; USB may still power Pico.

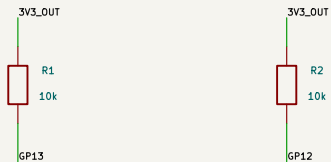
Stop playing before source changes: reboot may occur.



External charging only: cell/holder fit still pending.

ERC checks carrier wiring only. Cell PCM, module electronics and assembled fault/load/fit behavior remain unverified. See power sheet.

BUZZER RESET BIAS



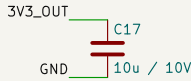
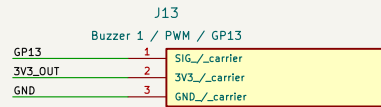
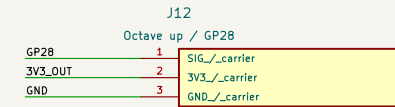
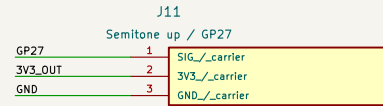
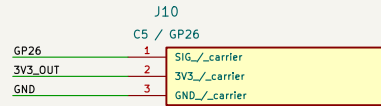
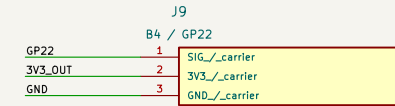
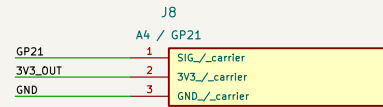
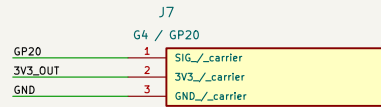
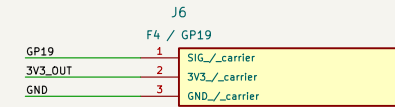
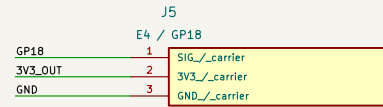
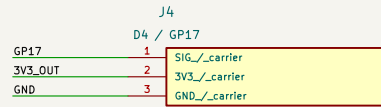
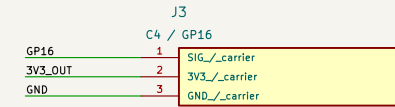
10k 1% each; nominal 0.66mA total when both LOW.

Carrier pull-ups require connected 3V3/GND/cables.

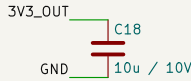
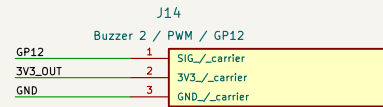
TEN TOUCH / TWO BUZZER CARRIER CABLE HEADERS

J3–J14 carrier–defined order: 1=SIG, 2=3V3, 3=GND. Map cable to module by signal label.

Harwin carrier header footprints assigned. No direct module–mount/hole geometry; see cable contract.



C17/J13: 10uF; retain C11 100nF. Place at buzzer header.



C18/J14: 10uF; retain C12 100nF. Place at buzzer header.

SIGNAL / POWER LIMITS TO RESOLVE

Touch: TTP223 at 3.3 V, direct mode (TOG=0). Firmware may adapt to confirmed active–high/low.

ST0238 low–level driver: R1/R2 bias GP13/12 HIGH at reset; firmware HIGH idle still required.

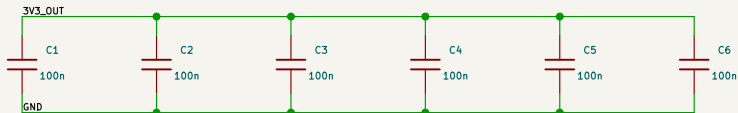
GP12 + GP13 share a PWM slice/frequency. Baseline firmware plays both at the same pitch.

Budget Pico 3V3_OUT current including 10 touch modules, 2 buzzer drivers and expansion loads.

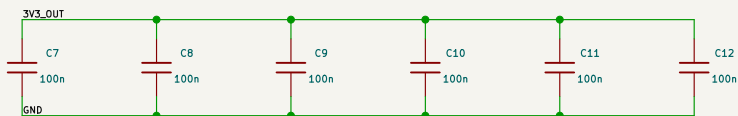
J1/J2: SSQ–120–01–G–S candidate assigned; mating fit, USB/BOOTSEL and battery clearances TBD.

Reference may be active–low (AHLB=1). Confirm actual polarity/pin order before firmware changes.

C1–C12: 100nF / 50V X7R per carrier header



C1/J3 .. C12/J14: place locally; short GND return.



Prototype review – fit and bench validation pending

Sheet: /
File: make_music.kicad_sch

Title: Make Music – provisional Pico carrier

Size: A3 Date: 2026–10–08

KiCad E.D.A. 10.0.6

Rev: P4

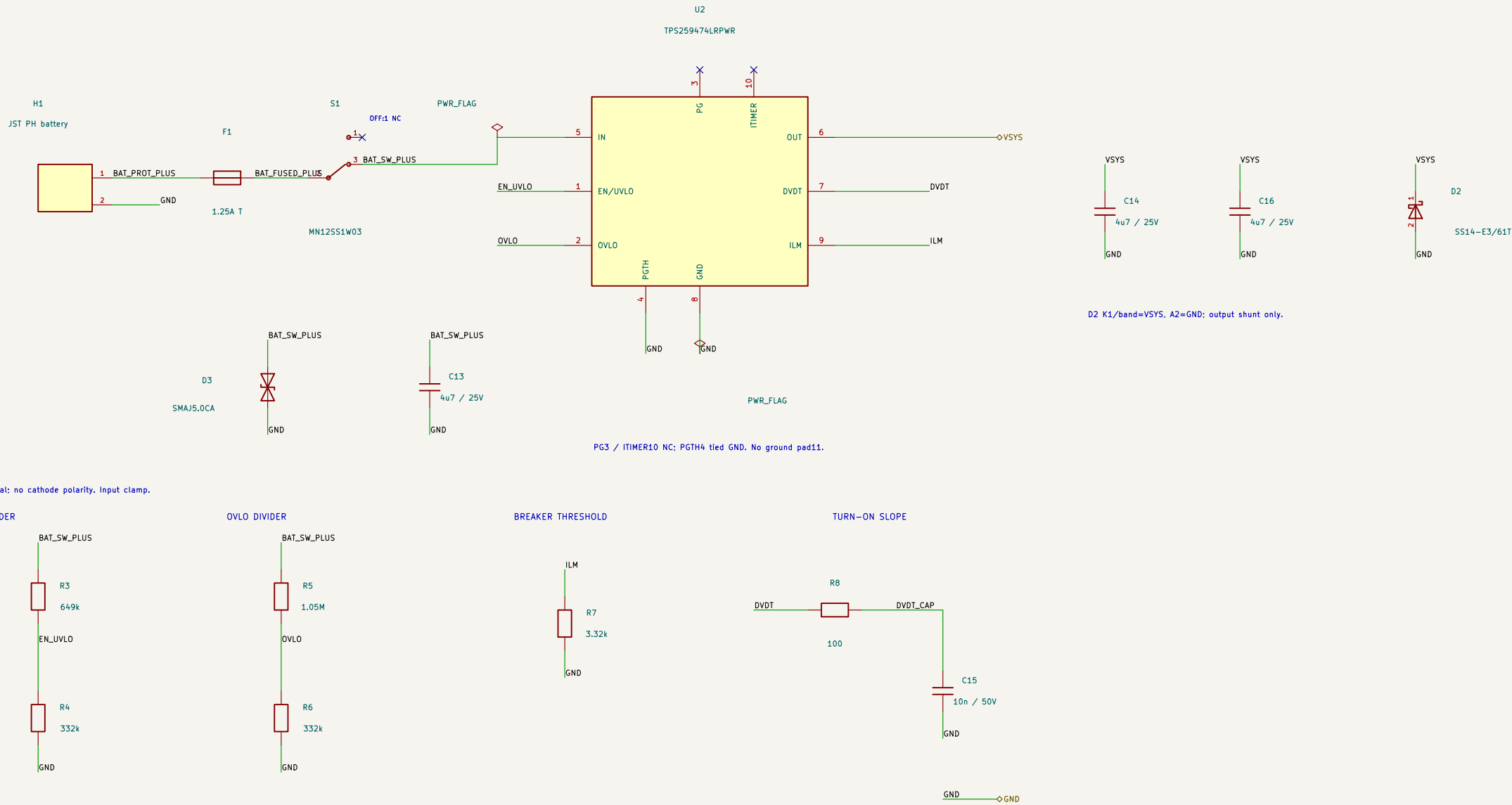
Id: 1/2

BATTERY BRANCH – FUSED, REVERSE-PROTECTED, EXTERNAL CHARGE ONLY

Prototype circuit from power PR #18. Physical fit, current, fault transients and source changes require bench validation.

ENTRY / BACKUP FUSE / SWITCH / LATCH-OFF eFUSE

LOCAL NONPOLAR CERAMICS / SHUNT CLAMPS



DESIGN LIMITS: branch $\leq 0.70A$; total3V3 $\leq 350mA$ including Pico / external $\leq 250mA$; ambient $0-40C$.

All VSYS capacitance including Pico $\leq 100\mu\text{F}$; effective output ceramics $>1\mu\text{F}$. No polymer hold-up parts.

UVLO nominal 3.221V off / 3.546V restart at 1N; breaker nominal 1.004A. Tolerance / 1S fault response need bench checks.

Stop playing before USB/battery changes; reboot may occur. S1 OFF / USB removal before service; external cell charging only.

F1 first at entry, before branches; 5mm-wide /35um copper approach lands per power contract. Keep unfused harness short/insulated.

Local C13 at IN/GND, C14/C16 at OUT/GND; short clamp/control loops. RPW IN5/OUT6 are large power contacts.

PWR_FLAGS declare external switched supply and return for ERC only; they do not add hardware or prove live power.

Reviewed candidate footprints and circuit are not fabrication approval. See [power-integration.md](#) and [power/branch-protection.md](#).

Prototype review – fit and bench validation pending

Sheet: /Battery protection/
File: battery_power.kicad_sch

Title: Make Music – fused battery power

Size: A3	Date: 2026-10-08
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Rev: P4

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